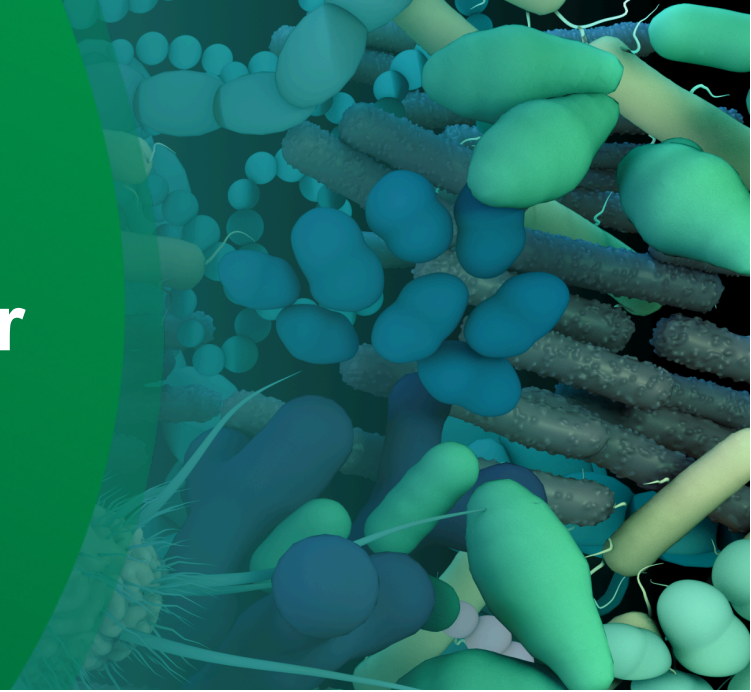
Intro to APIs and Python for data collection



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Data Science for Life Scientists course 2026

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Python. Why-thon?

- (Relatively) extremely readable
- Good for both *scripting* and *coding*
- Very extensive library and ecosystem
- Many possible paradigms:
  - Run scripts
  - Make command-line tools
  - Make web tools
  - Make GUI tools
  - Notebooks
  - Pipelines

Readable?

```
import pandas as pd
from matplotlib import pyplot as plt

dataframe = pd.read_excel("data.xlsx")
dataframe.drop_duplicates(subset=['patient_id'])

for country, patients in dataframe.groupby('country'):
    plt.scatter(patients['age'], patients['hba1c'], color='red')
    plt.title(country)
    plt.save(f"{country}_patients.png")
```

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Readable?




```
C     patients_pgplot.f (FORTRAN 77 style)
C     Input format per line: patient_id  age  hba1c  country
C     Example:
C       101 45 7.2 UK
C       102 53 6.8 UK
C       103 48 8.1 FR

      PROGRAM PATIENTS_PGPLOT
      INTEGER MAXN, MAXC
      PARAMETER (MAXN=10000, MAXC=200)

      INTEGER PID(MAXN), AGE(MAXN)
      REAL    HBA1C(MAXN)
      CHARACTER*32 COUNTRY(MAXN)
      INTEGER N, I, J, K

C     unique countries
      CHARACTER*32 UC(MAXC)
      INTEGER NUC

C     temp arrays per country for plotting
      REAL X(MAXN), Y(MAXN)
      INTEGER M

C     device name buffer
      CHARACTER*128 DEV
      CHARACTER*128 TTL
      INTEGER IER, L

C     read data
      N = 0
      OPEN(UNIT=10, FILE='data.txt', STATUS='OLD')
10    CONTINUE
         IF (N .GE. MAXN) GOTO 20
         READ(10,*,END=20) PID(N+1), AGE(N+1), HBA1C(N+1), COUNTRY(N+1)
         N = N + 1
         GOTO 10
20    CONTINUE
      CLOSE(10)
      IF (N .EQ. 0) THEN
         WRITE(*,*) 'No rows read.'
         STOP
      ENDIF

C     de-duplicate by patient_id (keep first occurrence)
      DO 40 I = 2, N
         IF (PID(I) .EQ. -1) GOTO 40
         DO 30 J = 1, I-1
            IF (PID(J) .NE. -1) THEN
               IF (PID(I) .EQ. PID(J)) THEN
                  PID(I) = -1
               ENDIF
            ENDIF
30       CONTINUE
40    CONTINUE

C     collect unique countries (only from kept rows)
      NUC = 0
      DO 80 I = 1, N
         IF (PID(I) .EQ. -1) GOTO 80
         IF (.NOT. IN_COUNTRY_LIST(COUNTRY(I), UC, NUC)) THEN
            IF (NUC .LT. MAXC) THEN
               NUC = NUC + 1
               UC(NUC) = COUNTRY(I)
            ENDIF
         ENDIF
80    CONTINUE

      IF (NUC .EQ. 0) THEN
         WRITE(*,*) 'No countries after de-dup.'
         STOP
      ENDIF

C     loop per country and plot a scatter
      DO 200 K = 1, NUC

C        gather this country's points
         M = 0
         DO 120 I = 1, N
            IF (PID(I) .EQ. -1) GOTO 120
            IF (COUNTRY(I) .EQ. UC(K)) THEN
               M = M + 1
               X(M) = REAL(AGE(I))
               Y(M) = HBA1C(I)
            ENDIF
120      CONTINUE

         IF (M .LE. 0) GOTO 200

C        compute simple axis ranges with small margins
         CALL MINMAX(X, M, XMIN, XMAX)
         CALL MINMAX(Y, M, YMIN, YMAX)
         CALL PADRANGE(XMIN, XMAX)
         CALL PADRANGE(YMIN, YMAX)

C        build device string; try PNG first (requires PNG driver in your PGPLOT)
         L = ITRIM(UC(K))
         DEV = UC(K)(1:L)//'_patients.png/PNG'

C        fall back idea (uncomment if PNG not available):
C        DEV = UC(K)(1:L)//'_patients.ps/PS'

C        open device (one panel)
         IER = PGBEG(0, DEV, 1, 1)
         IF (IER .NE. 1) THEN
            WRITE(*,*) 'PGBEG failed for ', UC(K)(1:L)
            GOTO 200
         ENDIF

C        set up plot box
         CALL PGENV(XMIN, XMAX, YMIN, YMAX, 0
         CALL PGLAB('Age', 'HbA1c', ' ')

C        title at top
         TTL = 'Country: '//UC(K)(1:L)
         CALL PGMTXT('T', 1.0, 0.5, 0.5, TTL)

C        points: red filled circles
         CALL PGSCI(2)
         CALL PGPT(M, X, Y, 17)

C        done
         CALL PGEND()

         WRITE(*,*) 'Wrote plot for ', UC(K)(

200   CONTINUE

      STOP
      END

C-----------------------------------------
C     helpers (F77-style)

      LOGICAL FUNCTION IN_COUNTRY_LIST(NAME,
      CHARACTER*(*) NAME
      CHARACTER*(*) LIST(*)
      INTEGER NLIST, I
      IN_COUNTRY_LIST = .FALSE.
      DO 10 I = 1, NLIST
         IF (NAME .EQ. LIST(I)) THEN
            IN_COUNTRY_LIST = .TRUE.
            RETURN
         ENDIF
10    CONTINUE
      RETURN
      END

      INTEGER FUNCTION ITRIM(S)
      CHARACTER*(*) S
      INTEGER I, L
      L = LEN(S)
      I = L
10    IF (I .GE. 1) THEN
         IF (S(I:I) .EQ. ' ') THEN
            I = I - 1
            GOTO 10
         ENDIF
      ENDIF
      IF (I .LT. 1) THEN
         ITRIM = 0
      ELSE
         ITRIM = I
      ENDIF
      RETURN
      END

      SUBROUTINE MINMAX(A, N, AMIN, AMAX)
      REAL A(*), AMIN, AMAX
      INTEGER N, I
      AMIN = A(1)
      AMAX = A(1)
```

Scripting, e.g.

Take some input and do bespoke thing

One-off convert a horrible data format

Call a couple of tools in a row

Coding, e.g.

Develop a library to perform a new algorithm

Run a web app

A CLI tool released to the community

Library and ecosystem

Standard Library (shipped with Python)

`json, toml, pickle, csv, sqlite`

`urllib`

`argparse`

Ecosystem(s)

`pandas, numpy, scipy`

`sklearn`

`requests, httpx`

`matplotlib, plotly`
`dash, flask, django`

Running scripts

```
python my_script.py
```

or

```
#!/bin/python
print("hello world")

chmod +x my_script.py
./my_script.py
```



Command-line tools

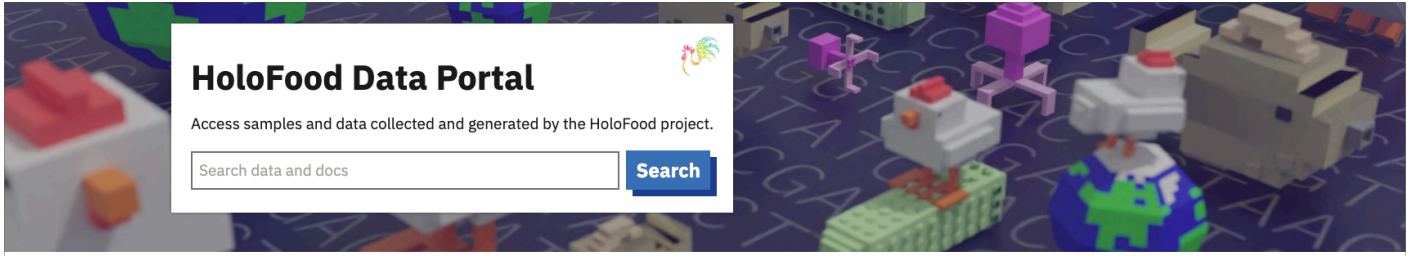
```
./my_tool --input data.csv --threshold 0.2 > output.csv
```

```
./interactive_tool
> What threshold to use?
0.2
```

```
some_software data.csv | my_tool --threshold 0.2 | other_software
```

Web tools

- Very popular non-Python alternative: **`R/Shiny`**
- Job-based: submit a file, run something, get result (**`Celery`**)
- Dataset-based (make some database usable to the world): **`Dash`**, **`Flask`**, **`Django`**
- Run Python actually in a browser (serverless): **`Pyodide`**



Home | Chicken samples | Salmon samples | Analysis summaries | API | Genomes | Viruses

Home > **Samples** Related: Animals

Filter samples

Animal system

Sample type

Accession contains

Title contains

Treatment Search

E.g. algae 2.0% "Day 60"

Animal accession contains

See the list of animals for details

Samples

Samples listed are extraction-level BioSamples, derived from animals across HoloFood trials.

Download all as TSV

Sample accession	Animal	Treatment	Treatment concentration	Trial timepoint	Sample type	Deta
SAMEA10104908	SAMEA112905066	Control	–	Day 21	metagenomic-assembly	Vie
SAMEA10104910	SAMEA112904813	Probiotic	–	Day 21	metagenomic-assembly	Vie
SAMEA10104911	SAMEA112904777	Probiotic	–	Day 35	metagenomic-assembly	Vie
SAMEA10104912	SAMEA112904752	Probiotic	–	Day 21	metagenomic-assembly	Vie
SAMEA10104914	SAMEA112904915	Control	–	Day 21	metagenomic-assembly	Vie
SAMEA10104915	SAMEA112904820	Control	–	Day 35	metagenomic-assembly	Vie
SAMEA10104916	SAMEA112904996	Prebiotic	–	Day 21	metagenomic-assembly	Vie
SAMEA10104917	SAMEA112905145	Prebiotic	–	Day 35	metagenomic-assembly	Vie

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GUI tools

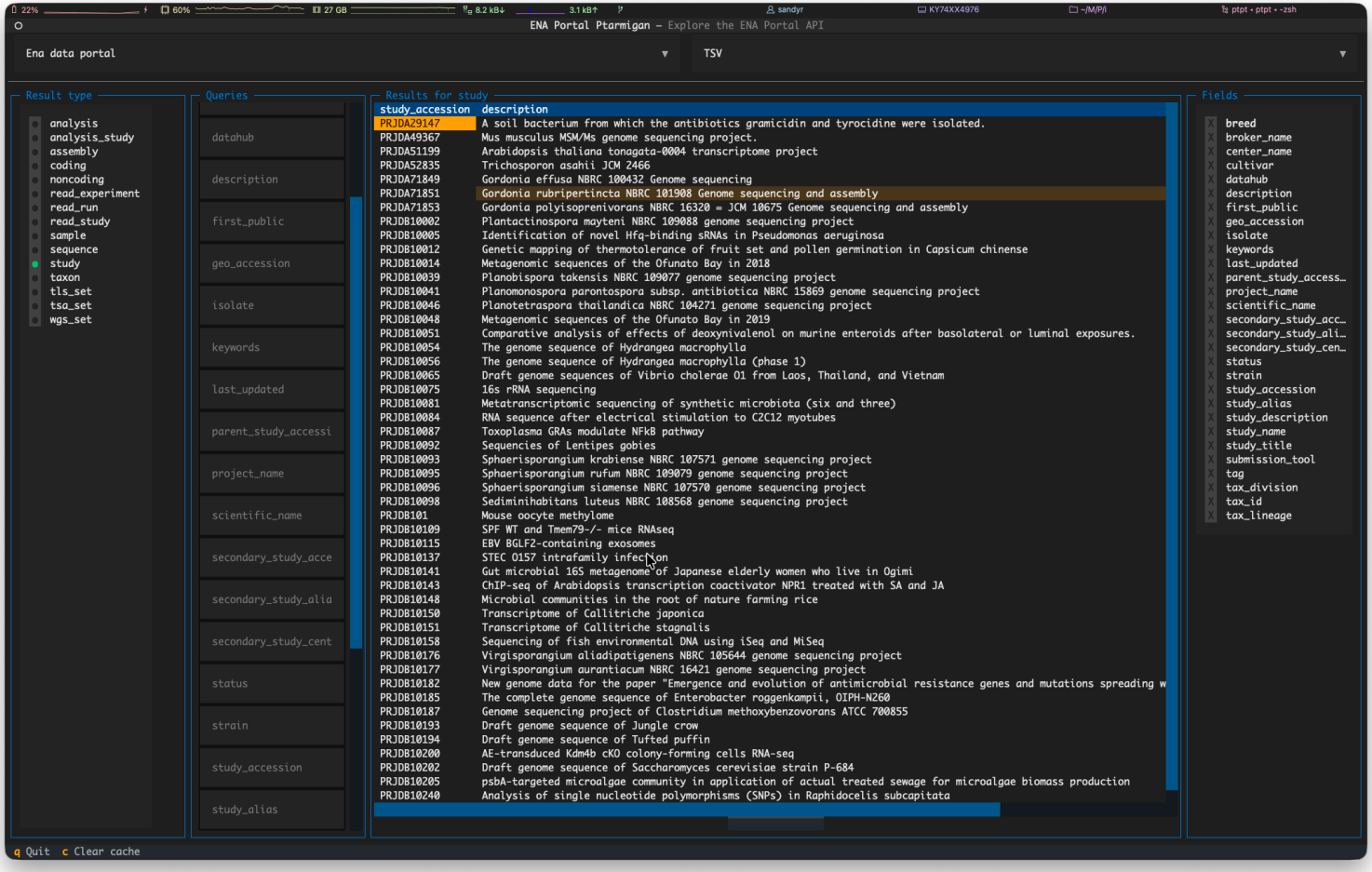
(Graphical User Interface)

- `tkinter`
- `qt`
- `guizero`

TUI tools

(Textual User Interface)

- `click`
- `textualize`



TUI

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https://github.com/SandyRogers/ena-portal-ptarmigan

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Notebooks



- Mostly used in the research and data-science spaces
- Increasingly broad tooling support
- **`Jupyter`** is pretty much the only Python choice
- Related: **`Quarto, Jupyter Book`** ("publish"/share analysis code as documentation)

Data handling

- If you learn one thing... make it **pandas**

```
# pip install pandas

import pandas as pd
df = pd.read_csv("my_data.csv")

df.dropna(subset=["age"])
df.sort_values(["age", "country"])
df.plot("age", "assessment")
```

Data visualisation

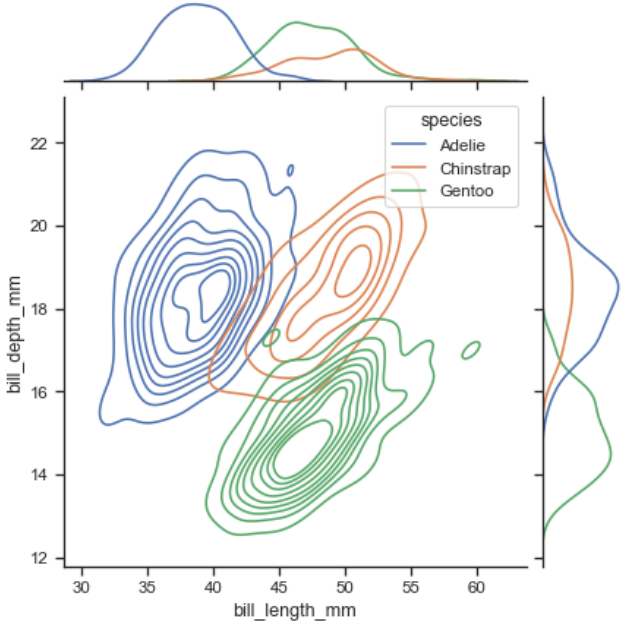
- If you learn one thing... make it **seaborn**

```python
# pip install seaborn

import seaborn as sns
sns.set_theme(style="ticks")

# Load the penguins dataset
penguins = sns.load_dataset("penguins")

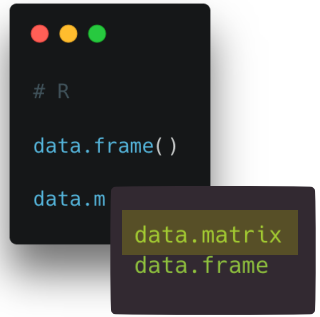
# Show the joint distribution using kernel density estimation
g = sns.jointplot(
    data=penguins,
    x="bill_length_mm", y="bill_depth_mm", hue="species",
    kind="kde",
)
```



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APIs: Application Programming Interfaces

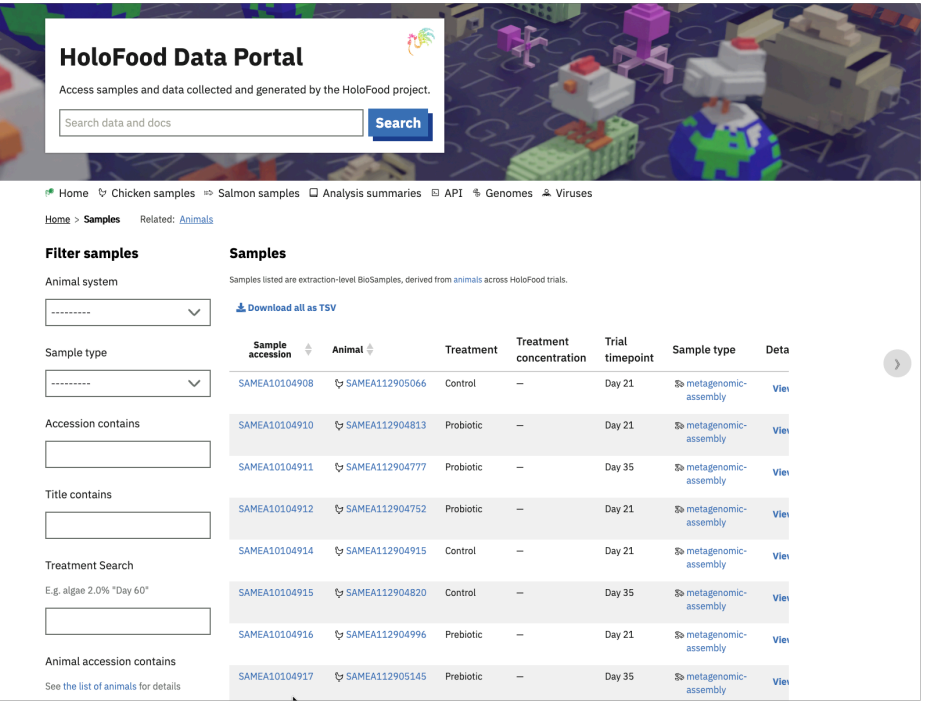
- Generically:
  "I, software X, do solemnly swear to understand the following commands".



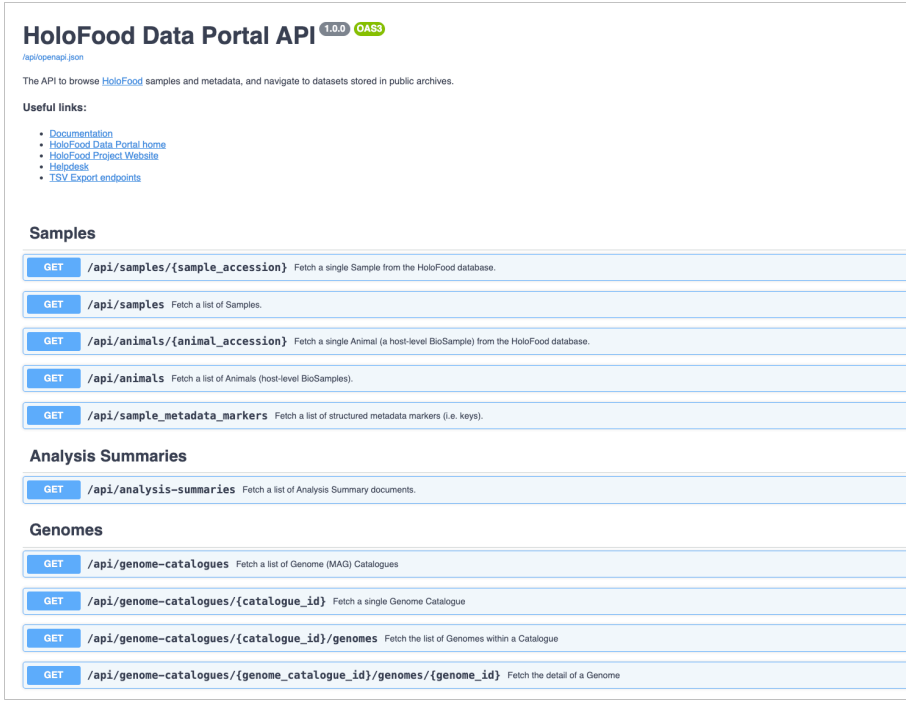
- Web specifically:
  "I, web service X, will give (and occasionally accept) data at the following URLs".

```
curl "https://api.open-meteo.com/v1/forecast?latitude=52.08&longitude=0.19&current=temperature_2m"
```

website



web api



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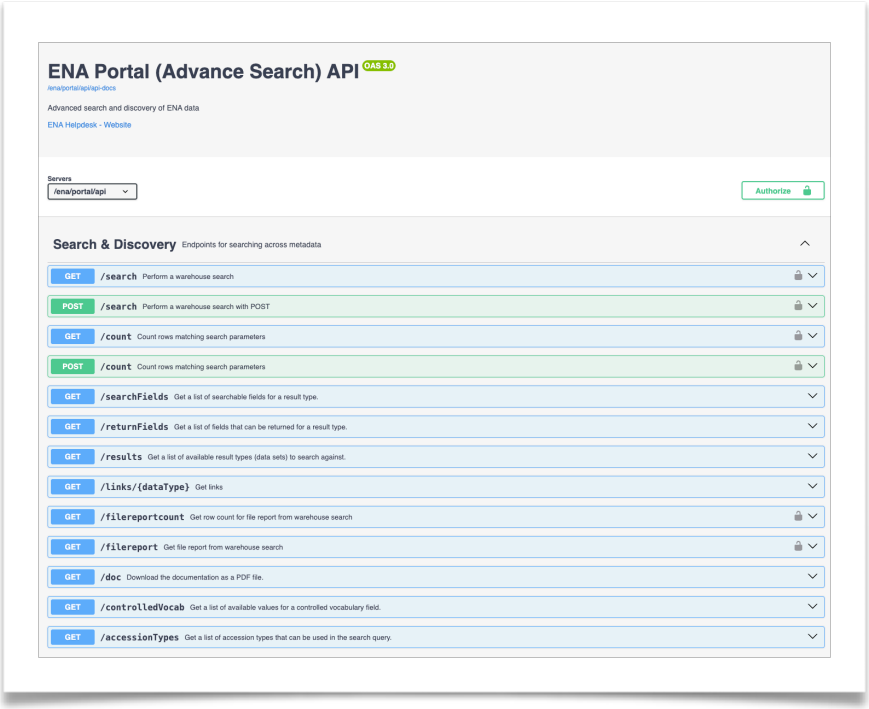
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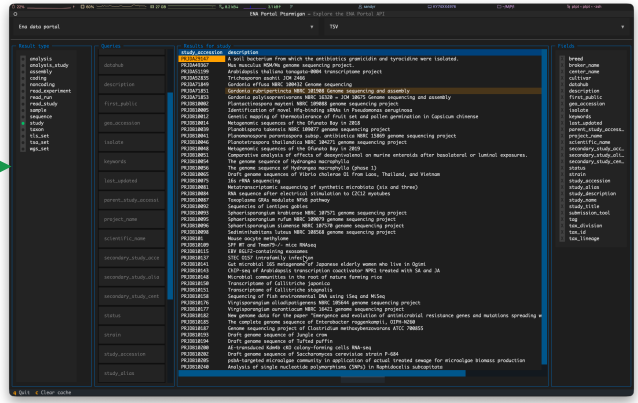
(A small selection of) EMBL-EBI's Web APIs

ENA European Nucleotide Archive

Programmatic access to nucleotide sequence archive



the API behind

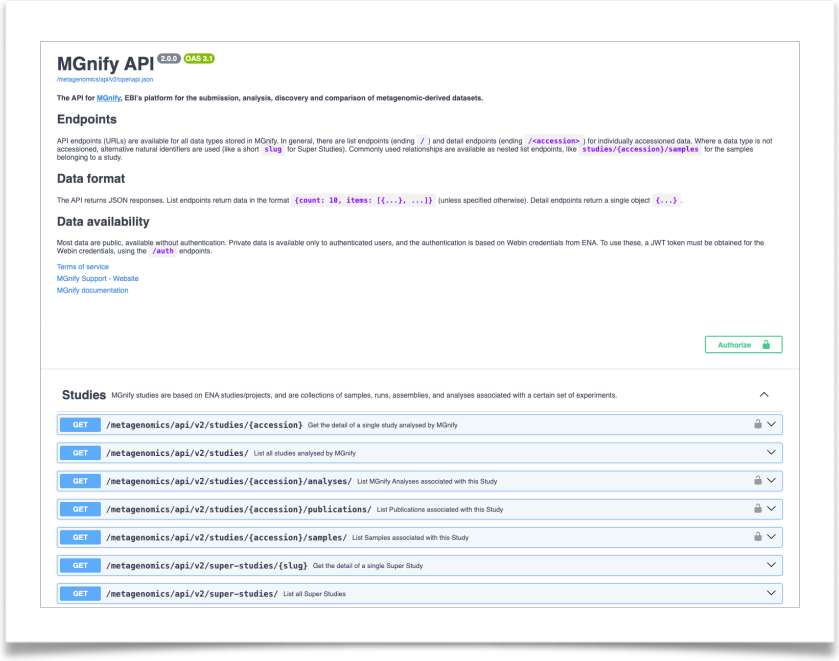


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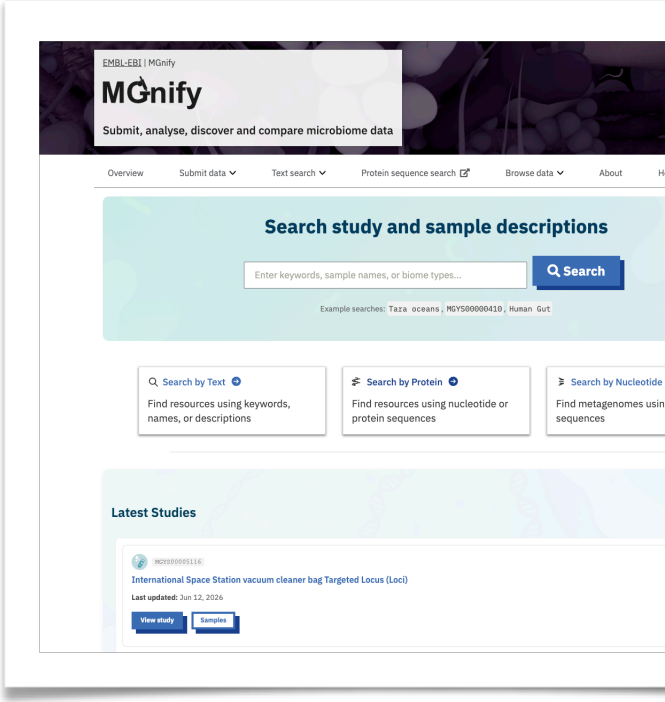
EMBL-EBI

(A small selection of) EMBL-EBI's Web APIs

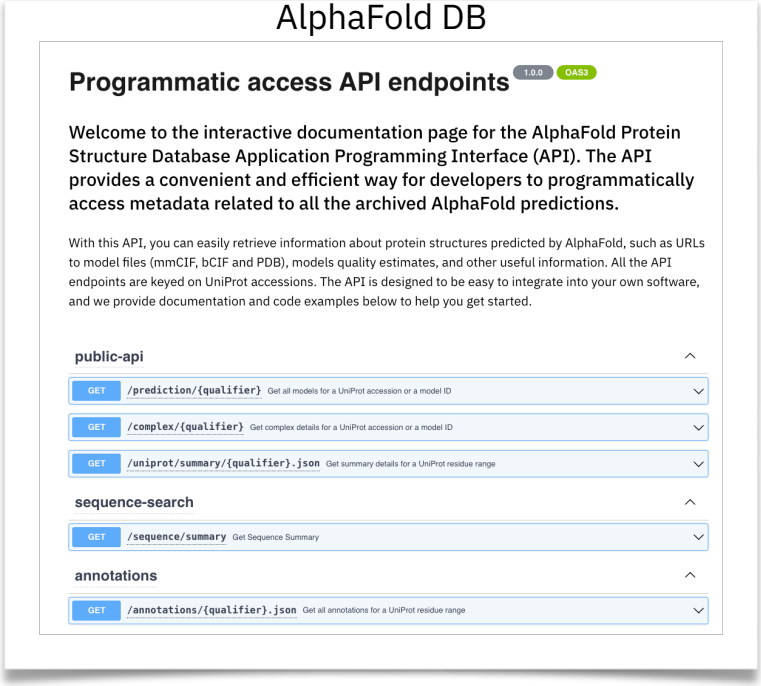
MGnify Programmatic access to metagenomic analyses



the API behind



(A small selection of) EMBL-EBI's Web APIs





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What can I do with these APIs?

- **Download data** without clicking on a website many times
- **Repetitive search** for and retrieval of meta/data for a list of identifiers/queries
- **Programatically submit** my data to a deposition archive, e.g. when a pipeline finishes
- Make a script that **checks a web service for new data** I might care about
- **Build myself a little tool** that augments my local data with some from the web

Practical, part 0

- Open VM
- Open a Terminal
-

```
curl "https://api.open-meteo.com/v1/forecast?latitude=52&longitude=0&current=temperature_2m" | jq
```

Practical, part 1

- Open VM
- Open a Terminal
-

```
cd course_dir/work_dir/Day_2

git clone https://github.com/SandyRogers/python-data-tutorials.git

cd python-data-turials/

conda activate sandy

pip install jupyter mgnipy

jupyter lab
```

- Open **`apis.ipynb`**

Practical, part 2

- Open VM
- Open a Terminal
-

```
cd course_dir/work_dir/Day_2

git clone https://github.com/SandyRogers/python-data-tutorials.git

cd python-data-turials/

conda activate sandy

pip install jupyter mgnipy

jupyter lab
```

- Open **`intro_to_data_handling_and_viz.ipynb`**

you probably did this two hours ago :)